Roll. No.: _____

Thapar Institute of Engineering and Technology, Patiala
Department of Computer Science and Engineering
EST (Summer Semester), July 2026

Course Name: **Quantum Computing** | Course Code: **UCS619**
Total Marks: **40** | Faculty: **NEG**
Total Time: **180 Minutes** | Date & Time: **17/07/26, 9 AM**

Note: Attempt **any 5** questions. Subparts of each question must be solved in one place.

1. **(a)** Another basis for $\mathbb{C}^2$ is (2)

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

Invert this relation to express $\{|0\rangle, |1\rangle\}$ in terms of $\{|+\rangle, |-\rangle\}$.

(b) A quantum system is in the state (3)

$$|\psi\rangle = \frac{3i\,|0\rangle + 4\,|1\rangle}{5}$$

1. Is the state normalized?
2. Express the state in the $|+\rangle, |-\rangle$ basis.

(c) Prove that $Z = iYX$ using matrix multiplication. (3)

2. **(a)** Design the circuit for correcting phase flip and bit flip using Shor's 9-qubit code. (4)

(b) Consider a Y error occured at second qubit. Explain the status of quantum system of each step of Shor's 9-qubit code. (4)

3. **(a)** Suppose that A is a projection operator in $\mathcal{H}_1$ where $A = |0\rangle\langle 0|$ and B is a projection operator in $\mathcal{H}_2$ where $B = |1\rangle\langle 1|$. Find $(A \otimes B)|\psi\rangle$ where

$$|\psi\rangle = \frac{|01\rangle + |10\rangle}{\sqrt{2}}$$

(4)

(b) A system is in the state

$$|\psi\rangle = \frac{1}{\sqrt{6}}|0\rangle + \sqrt{\frac{5}{6}}|1\rangle$$

A measurement is made with respect to the observable Y. What is the expectation or average value? (4)

4. **(a)** What do you mean by superdense coding? Explain its complete working with proper circuit diagram. (4)

(b)
1. Explain how BB84 quantum cryptography scheme secures the communication.
2. In B92 protocol, state the four rules based on which Bob decides his observation in $+$ basis or $\times$ basis.

(4)

5. **(a)** Consider a function with two inputs such that $f(x) = 1$. Explicitly show that the Deutsch-Jozsa algorithm works in this case by generating the vector $|y\rangle = |00\rangle$ as the final output. (4)

(b) Given the search space $|000\rangle, |001\rangle, \ldots, |111\rangle$:

1. Design the Grover's algorithm circuit (with the detailed oracle and Grover diffusion operator) to find the element $|100\rangle$ in the search space.
2. Explain the value at each point for searching the element $|100\rangle$.

(4)

6. (a) A three-qubit system is in the state (4)

$$|\psi\rangle = \left(\frac{\sqrt{2}+i}{\sqrt{20}}\right)|000\rangle + \frac{1}{\sqrt{2}}|001\rangle + \frac{1}{\sqrt{10}}|011\rangle + \frac{i}{2}|111\rangle$$

1. Is the state normalized? What is the probability that the system is found in the state $|000\rangle$ if all 3 qubits are measured?
2. What is the probability that a measurement on the first qubit only gives 0? What is the postmeasurement state of the system?

(b) What do you mean by entanglement? Discuss all four types of entangled Bell state. Also draw the quantum circuit to generate any one of the Bell states.

End of the Question Paper